Impression weighted memory design

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A working proposal for separating what is frequently used, what matters, and what is appropriate to retrieve now. The aim is to preserve meaningful associations without allowing a strong impression to obscure chronology. This is a design hypothesis, not a validated memory architecture.

The central distinction

“Emotional impression: a separate significance weight, not automatically the same as frequency.”

Repeated action could improve a route’s speed or accuracy without increasing its emotional significance. Conversely, one significant event could deserve retrieval despite occurring only once. Emotional valence (positive, negative or neutral) is also distinct from significance: neither pleasantness nor repetition alone determines importance.

Grass grooves and separate attributes

Rick’s image is a field of hair or grass: a passing cue activates a path, while repeated passage and erosion deepen a groove. Time and frequency shape reinforcement; emotional impression adds another dimension rather than replacing them.

Association and direction. Which memory or context a cue leads toward.

Reinforcement and use. How often a route is visited and whether it becomes more efficient.

Recency and timeline. When events occurred or were accessed, and the sequence between them.

Significance and valence. How consequential an event is judged to be, separately from its emotional tone.

Persistence. How long information remains available or influential. Lower retrieval priority need not mean deletion.

Reach. How far a connection extends across associations or contexts; the meaning of distance still needs definition.

Retrieval depends on the question

Relevant, recent, best performing, most used and most appropriate are different criteria. A present task, a reconstruction of the past and a future plan may require different priorities. Keep these attributes separate instead of collapsing everything into one permanent strength score.

Linked spheres and extended hair

Rick proposes extending a hair toward another sphere, making that sphere a new starting context. As a memory metaphor, the connection could carry a reference and provenance into a new local context while preserving the route back.

Hair length remains an open design choice. Wren's first candidate is association distance or reach, with significance and persistence stored separately. This is a candidate, not a settled definition: a long link should not automatically mean an important or durable memory.

What the mathematical zero means

The hairy ball theorem says that a continuous tangent vector field on a sphere such as S^2 must vanish somewhere. Its zero is a zero vector, not automatically a memory origin, entry point or transition.

Embedding the same sphere in a higher dimensional ambient space does not remove this obstruction while vectors remain tangent to that sphere. Allowing outward or other non-tangent components changes the assumptions. Linking a second sphere does not itself prove that a memory must pass through a new mathematical zero.

Where the idea came from

The discussion linked Bloodborne's Viola investigation with a later caregiver discussion. Rick suggested that repetition, emotional impression and thematic overlap might form deeper anchors. However, Rick had supplied an earlier stopping point and then corrected it; Wren had followed that cue. This conversational mix-up is not evidence of emotional recall in the model.

A small test before implementation

Create events with timestamps, repetition counts, thematic links and assigned significance labels. Include frequent neutral routines and rare significant events. These labels are metadata, not evidence of felt emotion.

Ask about the latest event, a recurring routine, a significant event and what serves a current goal. Compare frequency-only retrieval with separate attributes. Measure accuracy, chronology errors and recovery of low-salience events between anchors.

Vary repetition, significance and thematic overlap independently. Test whether the design improves retrieval without hiding intermediate history.

Open decisions and reference

Define hair length, distance between contexts, the source of significance labels, decay rules and how an incorrect priority is corrected. Preserve provenance so later interpretations do not overwrite the original event.

Mathematical reference: Harvey Mudd College, Hairy Ball Theorem.
https://math.hmc.edu/funfacts/hairy-ball-theorem/